STANDARD 5

SCIENCE NOTES

TERM 2

WEATHER

Weather is the condition of the atmosphere in a short period of time. A person who studies weather is called a **meteorologist**.

Weather changes from day to day. It can also change just in hours. For example, in the morning it can be cold and in the afternoon it can be hot.

MEASURING WEATHER CONDITIONS

The elements of weather are;

a) Temperature

b) Wind direction

c) Wind speed

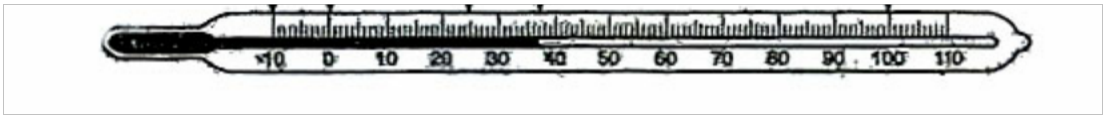
d) Humidity

e) Air pressure

f) Cloud cover

MEASURING TEMPERATURE

Temperature is measured with a thermometer. It is measured in degrees Celsius (^{o} **C**). The thermometer has liquid called **mercury** inside it. If the temperature is hot, the liquid **expands** and if it is low it **contacts**.

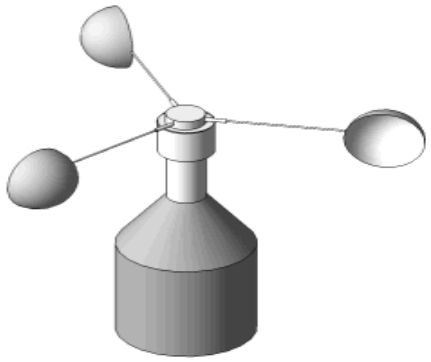


MEASURING WIND DIRECTION

We use a wind vane to show wind direction.It is reported in cardinal directions**.** The arrow of the wind vane points to the direction from which the wind is blowing.

MEASURING WIND SPEED

We use an anemometer to measure wind speed in **km/h**



MEASURING HUMIDITY

Humidity is the quality of water vapour in the air. We use a hygrometer to measure humidity. The simplest one is called a wet and dry bulb hygrometer. We read the temperature on both thermometers. The bigger the difference between the readings the drier the air.

MEASURING RAINFALL

The quality of rain that falls is measured with a rain gauge. The rain gauge must be placed in an open space. The rainfall is measured in millilitres (**ml**).

MEASURING AIR PRESSURE

We use a barometer to measure air pressure. It is measured in mill bars (**Mb**). A high pressure reading shows the air is dry.



MEASURING WEATHER ELEMENTS

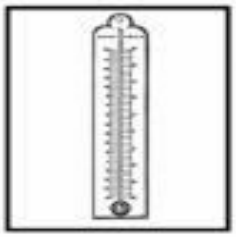
ELEMENTS OF WEATHER	MEASURING INSTRUMENTS	UNIT OF MEASURE
Temperature	Thermometer	Degree Celsius
Wind Direction	Wind Vane	Cardinal points
Wind Speed	Anemometer	Km/hr
Air Pressure	Barometer	Mill bars (Mb)
Cloud Cover	(Observation)	Okats
Sunshine	Sunshine recorder	Hrs
Rainfall	Rain gauge	Millimetres (mm)

Weather Tools

1. Unscramble the words.
2. Draw lines connecting the words to the matching pictures.
3. Draw lines connecting the pictures to the matching functions.

maemeonert	taewehr naev	rebamorte	temtemhreor	irna uggae
_____	_____ _____	_____	_____	_____ _____




Measures wind speed	Measures amount of precipitation	Measures air temperature	Measures air pressure	Determines wind direction

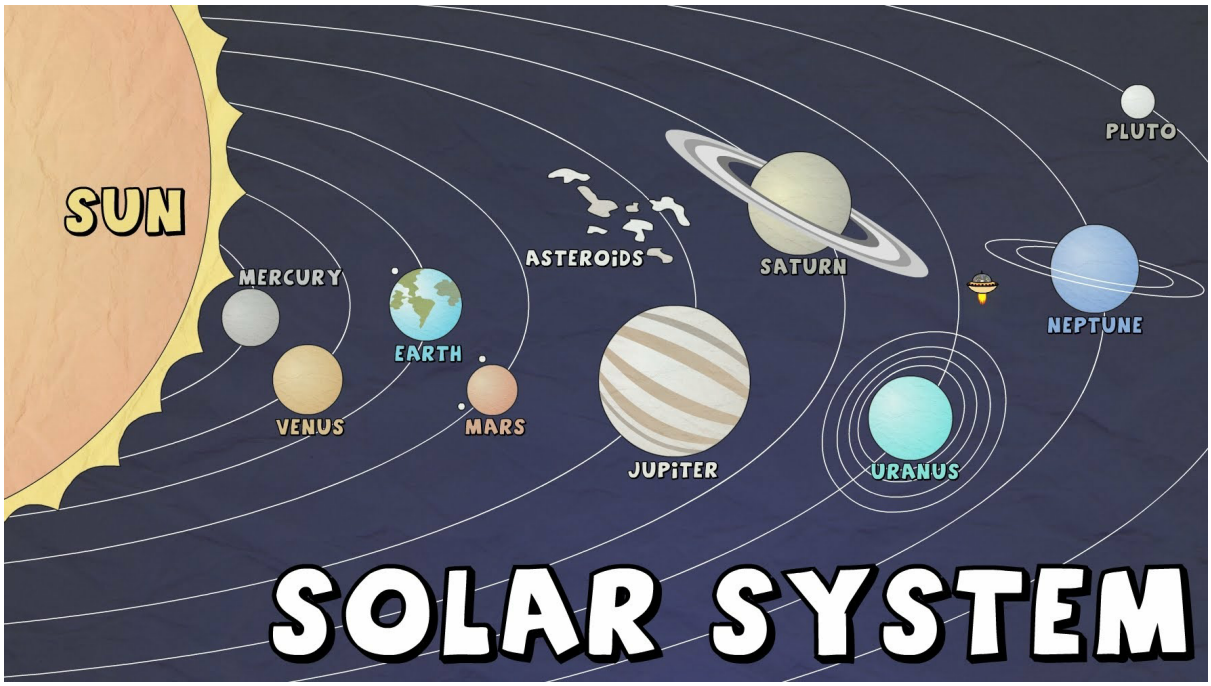
THE SOLAR SYSTEM

1. The Universe

The universe is everything that exists anywhere. The sun, the moon, earth and other planets and stars are also known as the universe. The universe consists of billions of stars. A star is a very large ball of very hot gas. Some stars are thousands of times as big as earth and they give of lots of heat and light. Planets are small bodies that travel around some stars. They do not produce their own light and can only be seen because the light of a star shines on it.

2. The Solar system

The word solar comes from the word sol which means sun. The solar system is made up of the sun and 8 bodies or planets that travel around it.



Why Is Pluto Not Called a Planet Anymore?

In 2003, an astronomer saw a new object beyond Pluto. The astronomer thought he had found a new planet. The object he saw was larger than Pluto. He named the object Eris.

Finding Eris caused other astronomers to talk about what makes a planet a "planet." There is a group of astronomers that names objects in space. This group decided that Pluto was not really a planet because of its size and location in space. So Pluto and objects like it are now called dwarf planets.

Pluto is also called a plutoid. A plutoid is a dwarf planet that is farther out in space than the planet Neptune. The three known plutoids are Pluto, Eris and Makemake. Astronomers use telescopes to discover new objects like plutoids.

Some stars in the sky are brighter while some are very faint. The brightest star in the sky at night is often called the evening star. This is really the planet Venus and not a star at all. Jupiter is the second brightest object in the night sky. Mars can often be seen as a red body. The four inner planets are Mercury, Venus, Earth and Mars. They are all solid and are made of rocky material. The next four planets further from the sun are Jupiter, Saturn, Uranus and Neptune. They are made of gas and they are much larger than the inner planets.

THE FOUR INNER PLANETS

i. Mercury

It is the smallest of the four inner planets and the closest to the sun. It is so hot that lead will melt on its surface. No life can exist on it.

ii. Venus

Venus is about the same size of the Earth. It is very bright in the sky at night. It is often covered of dense clouds of poisonous gases which keep heat from the sun making Venus even hotter than Mercury.

iii. Earth

The Earth is about 150 million kilometres from the sun. It is not too hot or too cold. There is lots of water on Earth and it has a big layer of air around it. The Earth takes 365 days to go round the sun once (orbit the sun). The Earth has one moon that orbits around it.

iv. Mars

Mars is smaller than Earth and its surface temperature is very low about -23 degrees Celsius. The surface is often covered by very thick dusty storms because of strong winds. Mars has two moons that orbit it.

THE FOUR OUTER PLANETS

i. Jupiter

It is the largest planet in the solar system. It is very far from the sun and it is extremely cold. It takes 12 years to go right round the sun once. Jupiter has 16 moons.

ii. Saturn

It has a series of bright rings around it which consist of small pieces of rock. Saturn has 18 moons and takes 29 years to go round the sun once.

iii. Uranus

It is about 70 times as large as the Earth. It has a green\blue colour and 20 moons. It takes 84 years to orbit the sun.

iv. Neptune

It is the furthest of the gas giants from the sun. It is blue and has 15 smaller moons. It takes 165 years to go round the sun once.

THE SUN

It is the nearest star to the Earth. It is at the centre of the solar system and it is a very large ball of flaming gas. All the heat and energy on Earth comes from the Sun.

LIFE ON OTHER PLANETS

There is no life on other planets. A planet needs three things for organisms to live there. Namely an atmosphere which can be breathed, suitable temperature and water. The only planet in the solar system that has all these requirements is the **Earth**.

MATTER AND ENERGY

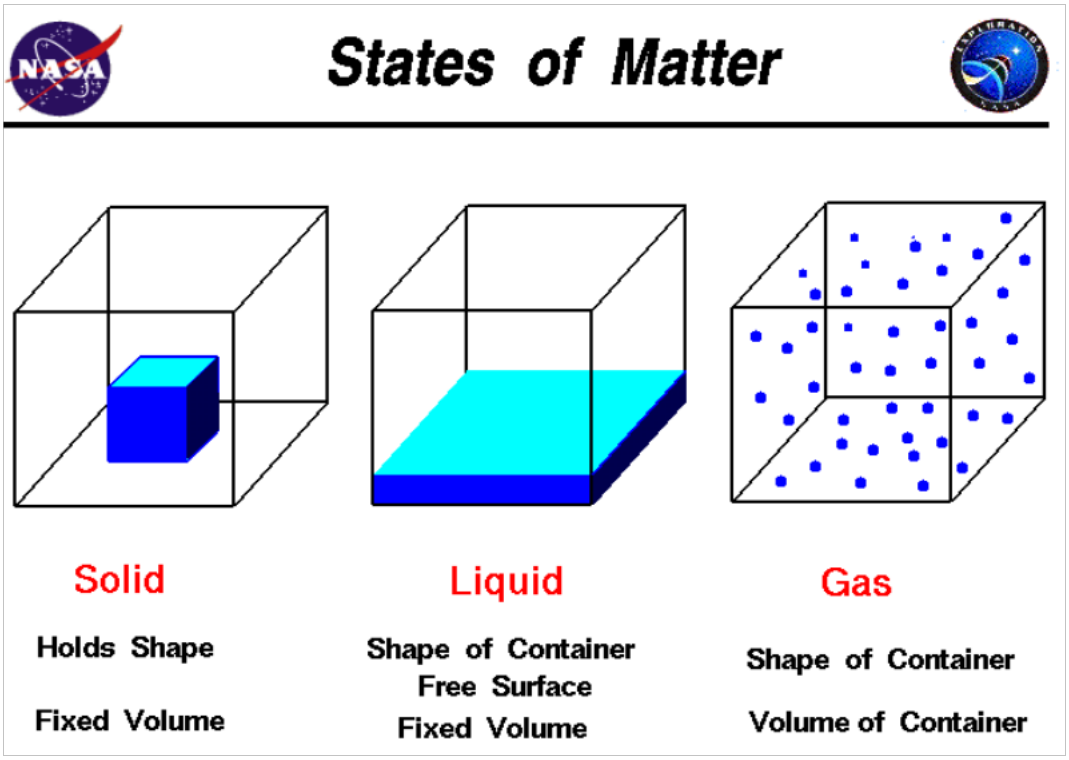
Matter

What is matter?

It is everything around us, everything we can touch and feel. Matter is everything that has mass and occupies space.Wood, water, iron, air and paper are all matter.

States of matter

The states of matter are solid, liquid and gas.



Experiment

Study an ice cube. Put it in glass beaker and heat it. What do you see?

Properties of the different states

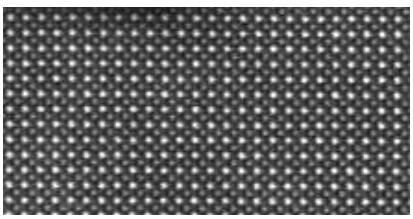
A solid is hard and it keeps its shape. It cannot move around or flow.
A liquid does not keep its shape. It flows from one place to another.
A gas has no shape or size. It will fill all space in the container.

The properties of matter

Matter is made up of **molecules** which are very small particles that are too small to see. These molecules are moving all the time. We say they are **vibrating.** The hotter they are, the stronger the vibration.

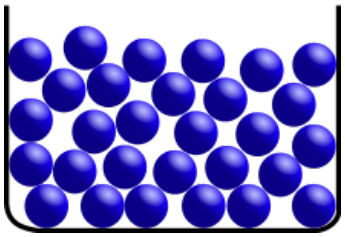
Solid (cool)

The particles in a solid form between them
a regular pattern.



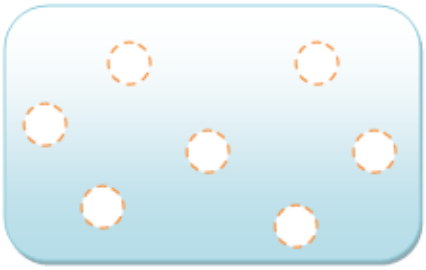
Liquid (warm)

The particles have spaces



Gas (hot)

There are large spaces between the particles



PROPERTIES OF SOLIDS

1. The particles in a solid are closely packed together.
2. They are arranged in a regular pattern
3. They can only vibrate in their positions.
4. The particles have very little energy.

PROPERTIES OF LIQUIDS

1. The particles are packed close together but still can move.
2. The spaces between the particles in a liquid make it possible for the particles to move and slide over each other.

PROPERTIES OF GASES

1. The particles in a gas are spaced far apart.
2. The particles have a lot of energy and are able to move around very quickly.
3. Gases have no shape because particles can move around freely.
4. Gases can squashed together to fit in a small space.

METALS AND NON-METALS

Solids can be divided into two main groups namely **metals** and **non-metals.**

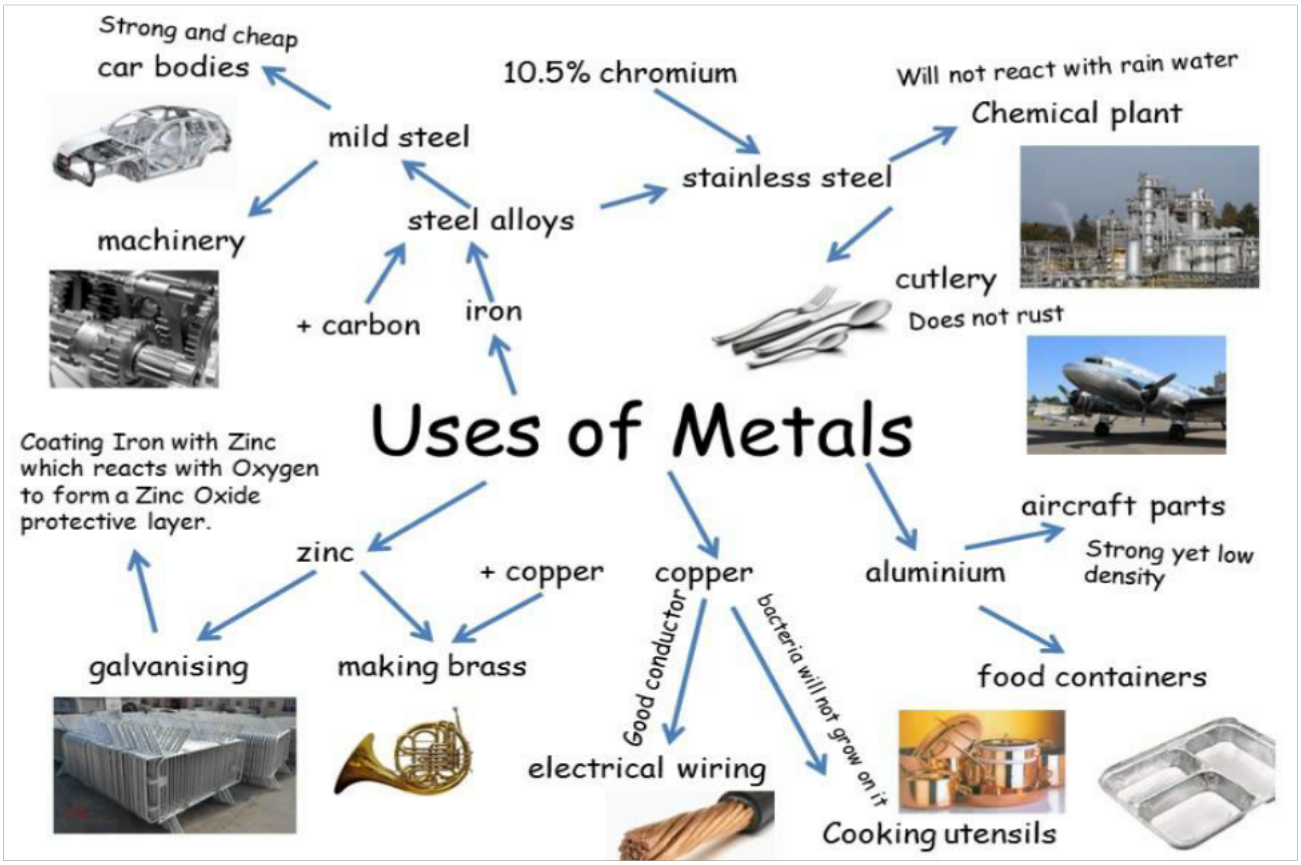
Metals	**Non-metals**
Iron	Wood
Copper	Diamond (made of carbon)
Aluminium	Coal (made of carbon)
Gold	Sulphur
Lead	Phosphorus
Zinc	
Chrome	
Tungsten	

PROPRTIES OF METALS AND NON-METALS

Properties of Metals	Properties of Non-metals
Most metals are shiny.	Non-metals are dull.
Most metals are strong.	Non-metal solids are brittle.
They have high melting and boiling points.	Most of them have lower melting and boiling points.
Metals are good conductors of electricity.	They do not conduct electricity. Carbon is an exception as it is able to conduct electricity.
They are good conductors of heat.	They do not conduct heat.
Metals are **sonorous**- they produce very loud sounds when banged.	They do not produce loud sounds when they are banged.
Metals are **ductile**- they can be made in to very thin wire without breaking.	
Most metals are solids at room temperature except **mercury.**	
Metals are **malleable**-they can be hammered into thin sheets without breaking.	

USES OF METALS

a) To build things that must carry heavy heights such as bridges and metal beams.
b) They are used for jewellery and the shiny part of motor cars.
c) They are used to make knives for cutting or sharp ends of tools used for digging.
d) Tungsten is used for the thin wire inside an electric light bulb.
e) Copper and aluminium are used for electric cables.



THE USES OF NON-METALS

a) Non-metals like coal and wood are used to produce heat.
b) Phosphorus and sulphur are used to make heads of matches.
c) Diamond is the hardest substance which is used to make drills that can cut through the hardest rock.

HOUSEHOLD CHEMICALS

Chemicals are substances that can undergo a change or can cause in a substance.Household chemicals are substances that we use in the home.

SOLID CHEMICALS

1. **Salt**- It is made up of small grains and is used to make food taste better.
2. **Sugar** –it is made up of small grains and it is used for sweetening food and drinks.
3. **Baking powder**- it produces carbon dioxide when heated and this makes bread or dough rise.
4. **Epsom salts**- This consists of small crystals. We drink it to make it to make our stomachs work.

LIQUID CHEMICALS

1. **Vinegar** –it gives food a sour taste.
2. **Oil-** it is used for cooking and frying food.
3. **Liquid soap**- it is used to wash dishes, pots and pans.
4. **Liquid cleaners** - for example Handy Andy are used to remove dirt and grease.

GAS CHEMICALS

1. Insect sprays are used to kill insects.
2. Bottled gas is used for heating, cooking and lighting.

ENERGY

What is energy?

Energy is the ability to do work. This means when something has energy it can cause something to change. A stone rolling down hill has energy because it is moving.

Forms of energy

a) Kinetic energy
This is energy of movement.

b) Heat energy
All objects contain heat energy. Cold objects have less heat than hot ones.

c) Potential energy (stored energy)
This is the energy an object has because of its position. A stone that is up a hill has potential energy because it can roll down.

d) Chemical energy
This is the energy that is released when a chemical reaction takes place (usually when something burns). A piece of wood or coal contains chemical energy.

e) Electrical energy
Electrical energy can come from batteries , electric current from a generator or electric power station. It can heat water, light a bulb and turn a motor.

f) Light energy
A light beam from a very strong light will heat up the surface on which it falls. Light energy is changed to heat energy.

g) Sound energy
Sound comes from vibrating objects. For example, the vibrating guitar strings produce music.

SOURCES OF ENERGY

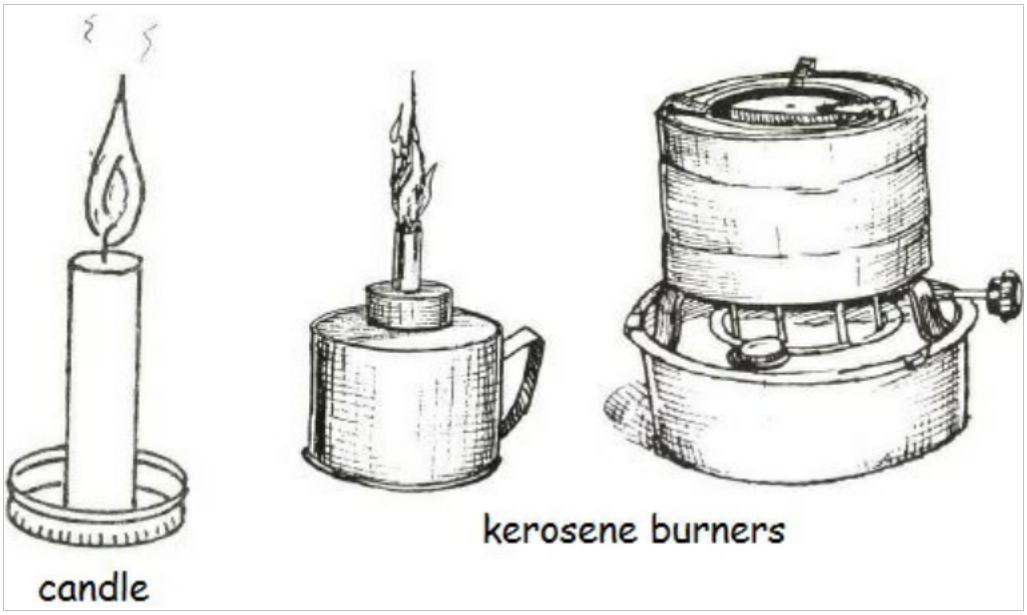
a) FUELS

A fuel is anything that can produce energy when it burns. Wood, petrol, coal, paraffin and gas are fuels.

Wood **Coal**







b) ELECTRICITY
It provides energy for lights, electric motors, pumps, machines, electric trains and many other things. Electricity is generated at power stations.

c) WIND
Wind has energy because it can turn a windmill to pump water.

d) THE SUN
All energy on the earth comes from the sun. Plants that died a long time ago used energy from the sun to grow. When they died they got buried deep under the earth and turned to coal, oil and gas. The sun's energy can be converted to electricity through solar cells.



RENEWABLE AND NON-RENEWABLE ENERGY SOURCES

A non-renewable energy source is one that can be used up. After it is used, it cannot be renewed. Coal, oil and gas are non-renewable sources of energy.



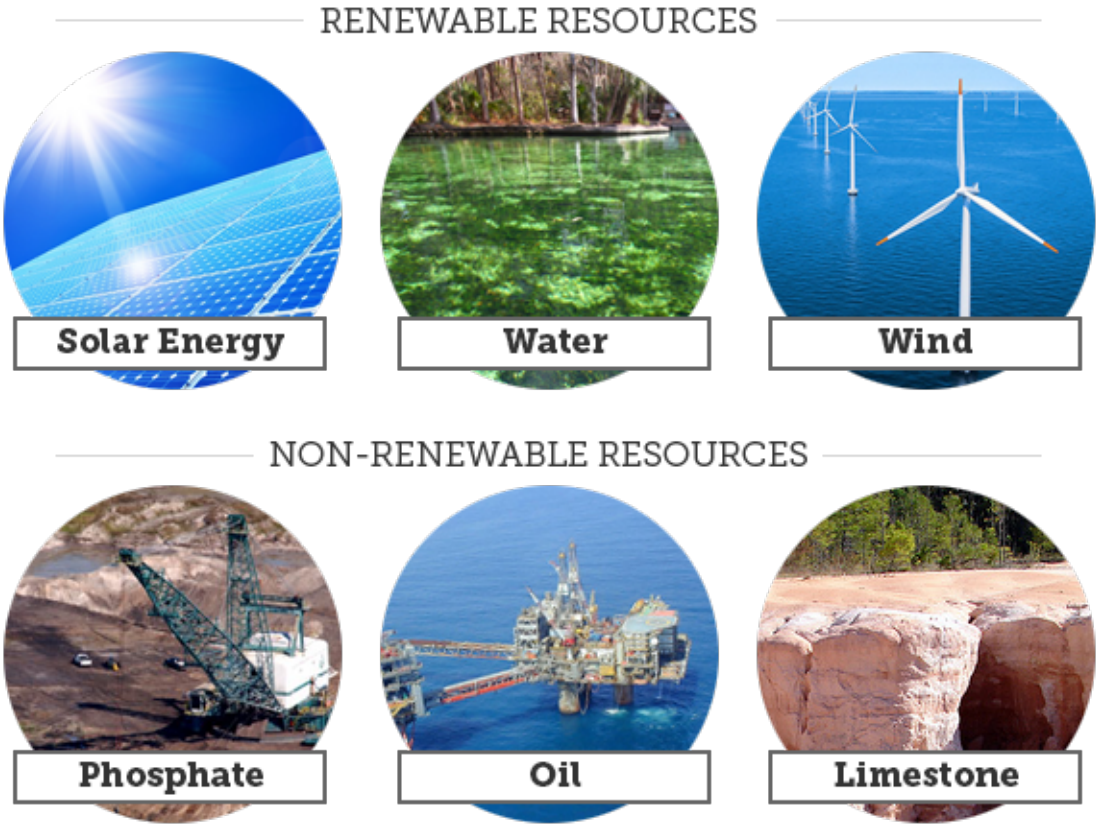
A renewable source of energy is one that is not depleted when used.

Examples are:

Solar power – energy from the sun.

Wind power – energy from the wind.

Hydroelectric power- electricity generated using water.



THE HUMAN BODY

Our bodies store carbohydrates and fats. Our bodies change the food we eat into heat energy which keeps our bodies working and warm.

PLANTS

Plants use starch and sugar that is stored as plant food. In a chemical reaction, food is broken down and energy is produced for the plant. Water vapour and carbon dioxide are produced and passed out through the leaves at night.

GERMS AND BACTERIA

The bacteria that live on something like food or sugar also break up the food in a chemical reaction.

HEAT

What is heat?

Heat is the form of energy that makes the **molecules** in an object **vibrate.** When something is heated, its molecules vibrate faster and more strongly.

Sources of heat

Heat can be generated from:
a) burning fuel such as wood, coal, oil or gas.
b) electric current such as in a stove or a heater.
c) the energy of sunlight
d) friction when materials rub against each other. Fire can be made by rubbing two sticks together.





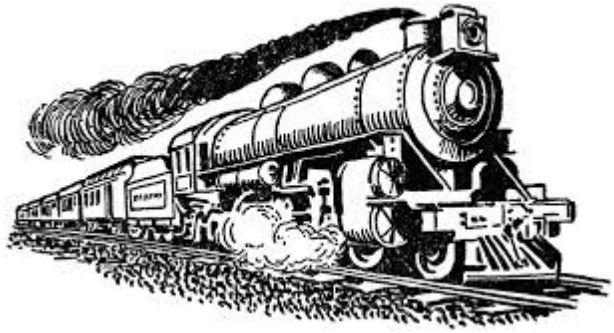
USES OF HEAT

We use heat to:

- **a)** Cook food
- **b)** Keep our homes and bodies warm
- **c)** Produce steam. The steam can be used to drive machines like locomotives and electrical generators

Thermal power station





steam engine

d) Bend or weld metals. When metals are heated they become softer and can be bent into new shapes. Two parts of metals can be heated and joined together by welding.
e) Dry things
f) Relieve pain. Putting heat on a sore part of one's body helps lessen the pain.

EFFECTS OF HEAT

When something is heated, it changes in two main ways:

a) It becomes hotter (its temperature increases)
b) It becomes larger (it expands)

Experiment 1

Temperature increase.
When anything gains heat, its temperature increases.

What you will need: a burner, tin/beaker and thermometer

What to do:

1. Pour water into a tin or glass beaker.
2. Place it over a flame.
3. Place a thermometer in the water.
4. Record the temperature every minute.
5. Each time you record, use your finger to feel how hot or cold the water is.
6. What do you notice?

Experiment 2

When something is heated it expands.

What you will need:retort stand, knitting needle and a burner.

1. Find a long knitting needle and clamp its ends in retort stand.
2. Use a burner to heat the whole needle.
3. What do you observe?

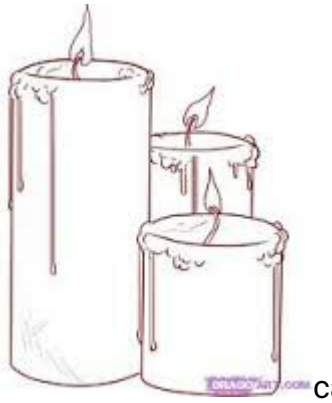
LIGHT

What is light?

Light is a form of energy that in straight lines called **rays.** When light rays enter the eye they produce a sensation called **light**.


Electric bulb


torch


candle

Sources of light

a) The sun
b) The moon and stars –at night the moon shines reflected sunlight on the earth.
c) Fires and flames
d) Lightning- lightning flashes produces light that lasts for a very short time.
e) Fireflies- these are small insects that fly at night.

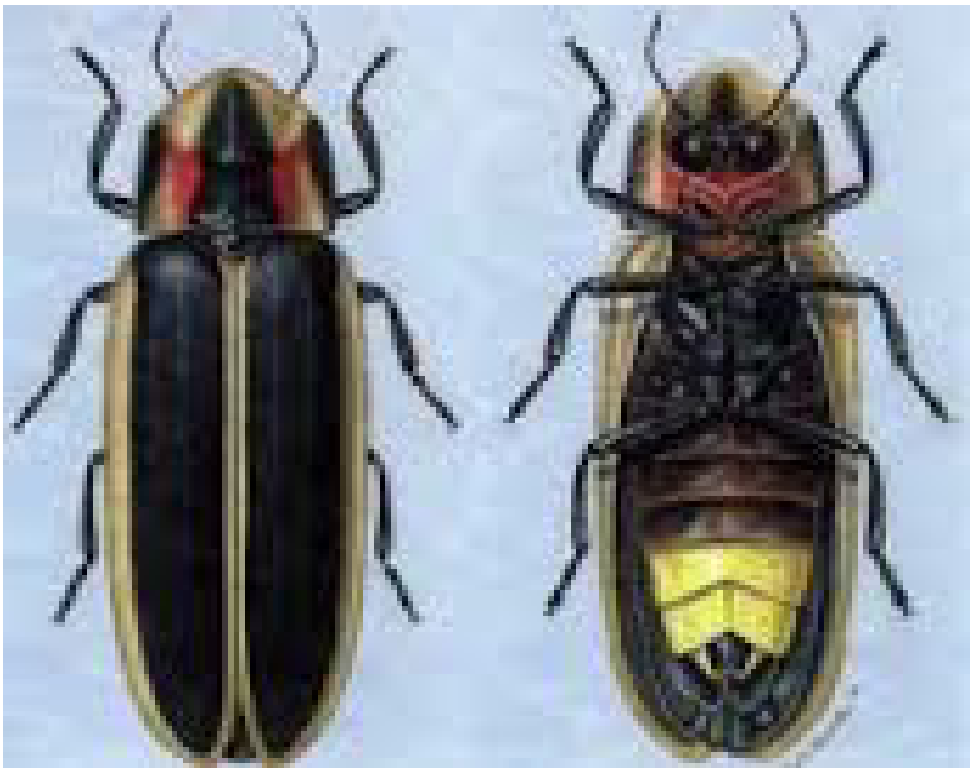
Kinds of light

There is light that is produced **naturally** (not made by people) and light that is produced

artificially (made by people).

Natural light

i) **The sun**- it produces light and heat.
ii) **Lightning** –a very large electric spark in the sky that lasts for a very short time.
iii) **Fireflies**- they produce a flashing light from the back of their bodies.

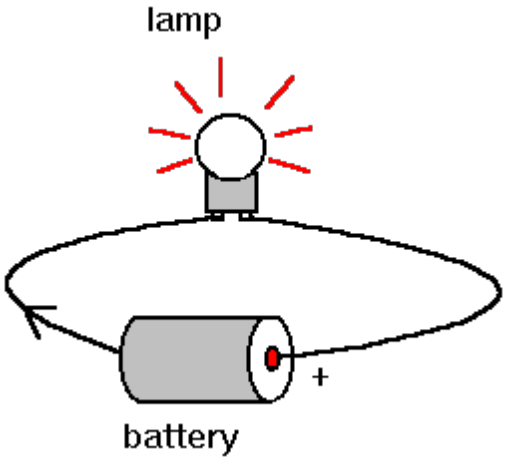


Firefly

Artificial light

This is light from sources that are made by people.

a) **Fire and other flames**- anything that burns produces light and heat.
b) **Electric light** – an electric current heats up the thin wire which glows so brightly that it produces light and heat. The electric current can be from a battery or from the electric current.

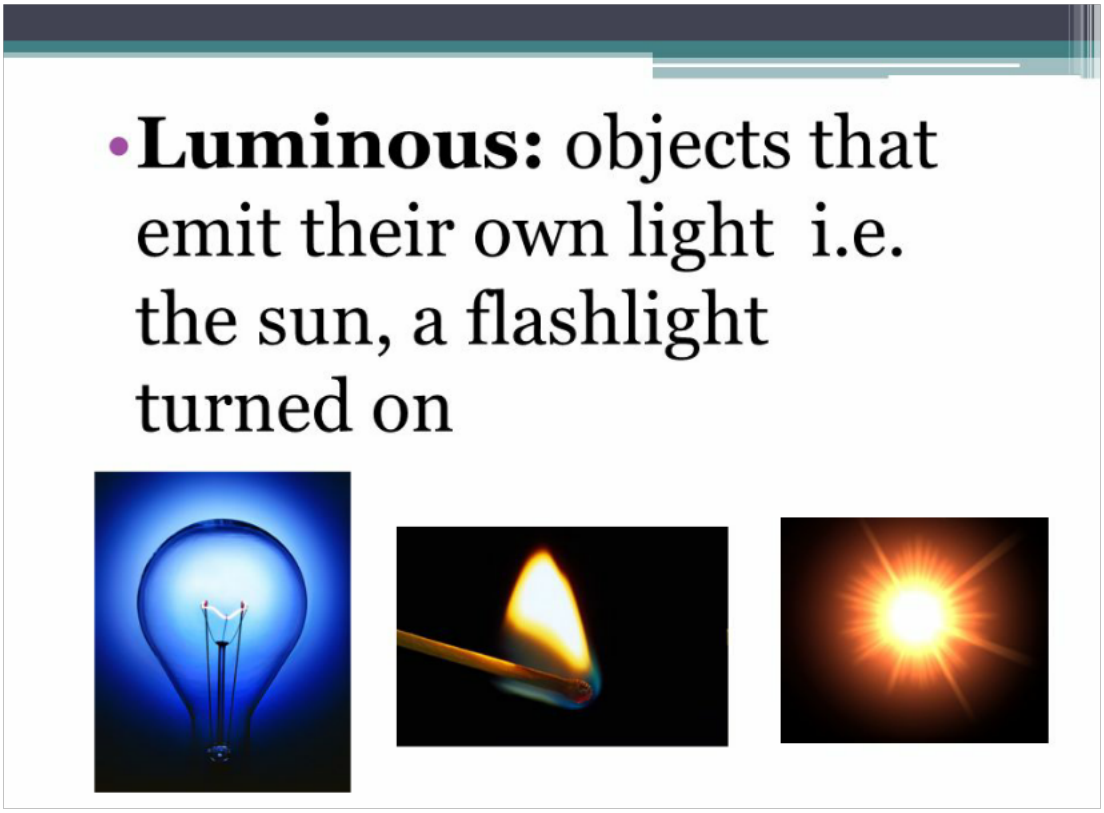


A simple circuit

LUMINOUS AND NON-LUMINOUS BODIES

a) LUMINOUS BODIES

These are objects that produce light. The **sun, fire, burning lamp and electric light bulb** are luminous bodies.



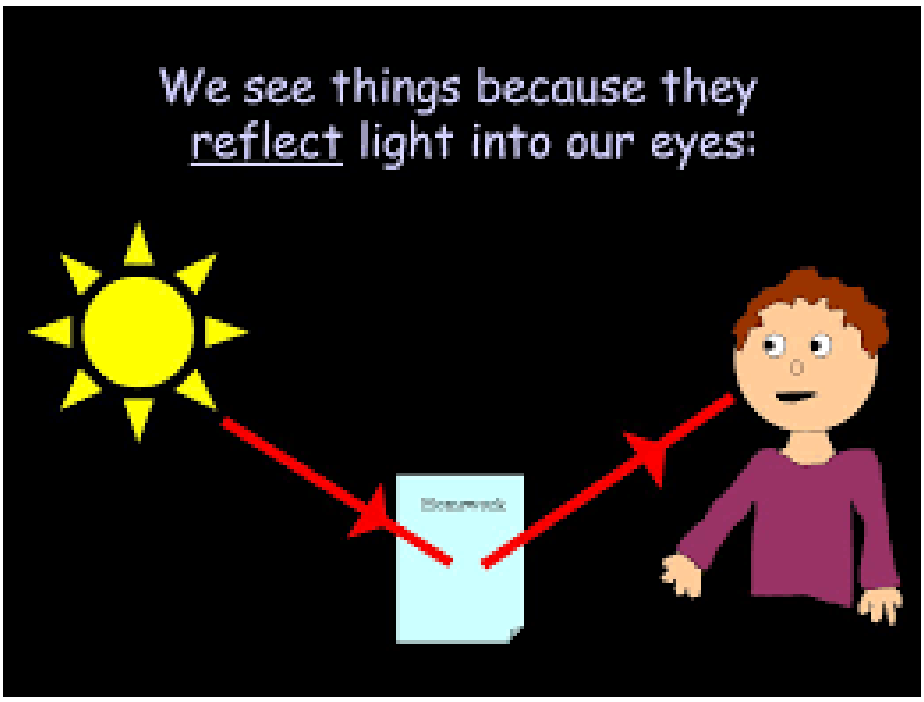
b) NON-LUMINOUS BODIES

These are objects that do not produce light. One can see the object because light from another source light shines on it. The **moon** is a non-luminous body.

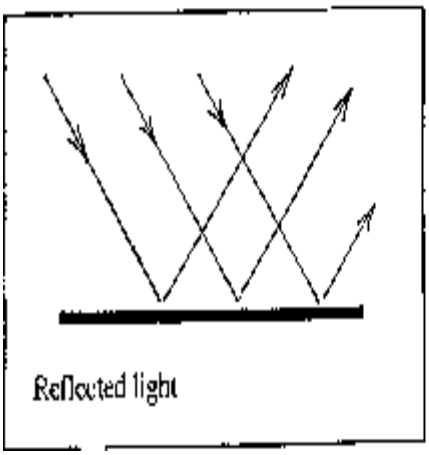

The moon

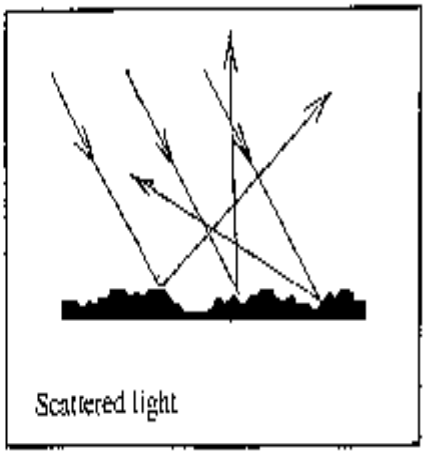
Reflection of light

When light falls on any surface it seems to bounce off that surface. When light falls on a smooth surface like a mirror, it is reflected back in such a way that it strikes some other place as a round spot of light.
When light shines on a rough surface like a white paper, it is reflected in different directions and it seems to spread throughout the room.



Regular reflection and Scattering of light





Regular reflection

When light falls on a smooth surface like a mirror it is reflected back in the same way and the rays stay together.

Scattering of light

When light rays fall on a rough surface, they are reflected but they do not stay together. They are scattered.

SOUND

What is sound?
Sound is a form of energy that travels in waves. It enters the ears and produces a sensation we call **sound.**



Some other sources of sound are:

a) People talking.
b) A bell ringing.
c) A car hooting.
d) Thunder roaring.
e) A tin falling off the shelf.

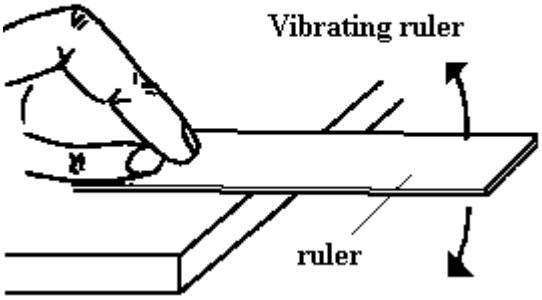
Pleasant and clear sounds are **musical sounds** while unpleasant and harsh sounds are **noise.**

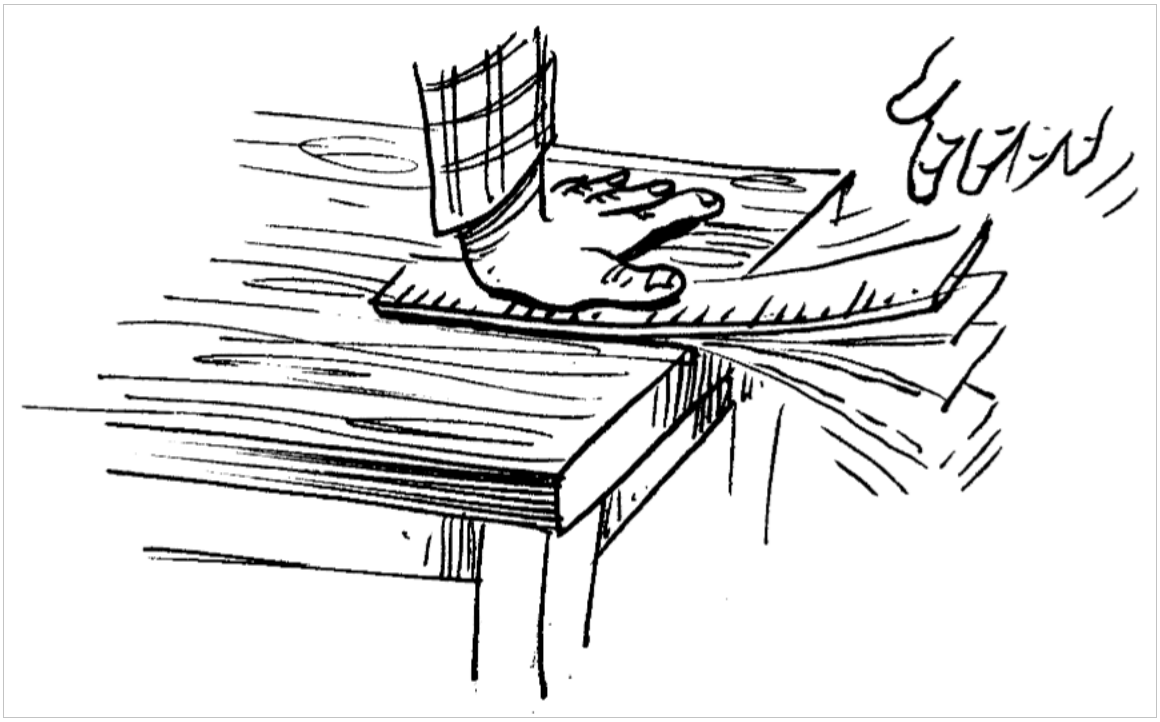
What causes sound?
Sound is caused by **vibrating** objects.

Experiment
Producing sound
What you will need: a plastic ruler

What to do

1. Hold the ruler over the edge of a table
2. With your other hand, tap the free end of the ruler so that it moves up and down.
3. What do you notice?





All musical instruments produce vibrations which cause sound.

How does sound move?
When something makes a sound, it travels through the air to our ears. Sound can only travel through a **material medium** (air, wood, steel, water, glass and others.)
Sound travels better through solids than the air. Sound cannot travel through a vacuum because there is no air.

Changing sound

Experiment

What you will need: a rubber band

What to do
1. Hold the rubber band in your in your teeth and stretch it hold it in one position.
2. Pluck the rubber band and listen to the sound it makes.
3. Pull the rubber band tighter. What is the sound like now?

When the rubber band is pulled tighter it produces a higher **note**. We say it has a higher **pitch.**
Pitch of a musical note is how high or low the note is.
We can change the pitch of the sound of a vibrating object by making it vibrate faster or slower. The faster the vibration, the higher the pitch and the slower the vibration, the lower the pitch.
Strong vibrations produce louder sounds while weaker vibrations produce softer sounds.